

C976

Science Methods—Secondary Biology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

In a room with a new fume hood, students claim that yeast makes more CO₂ at 35°C than at 20°C. Their balloons measured 18 cm around at 35°C and 9 cm at 20°C. Which statement is the reasoning?

- A. The 35°C balloon was twice as large as the 20°C balloon.
- B. Yeast always makes more gas at any higher temperature.
- C. Enzymes in yeast work faster at warmer temperatures up to an optimum, so more sugar is broken down and more CO₂ released.
- D. Yeast makes more CO₂ when it is warm.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A student writes: Claim, the plant grew toward the light. Evidence, the stems bent 40° toward the window. Reasoning, “because the stems bent toward the window.” What feedback is most useful?

- A. Collect more data points before writing any reasoning.
- B. Measure the angle to the nearest 0.1°.
- C. Swap your claim and evidence.
- D. Your reasoning repeats the evidence; explain the principle, such as growth hormone building up on the shaded side so those cells lengthen.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A teacher prints a DNA model on a 3-D printer bought with a district grant. Which use best reflects the practice of developing and using models?

- A. Students use the model to predict how a base change affects pairing, then list what the model cannot show.
- B. Students treat the model as an exact copy of real DNA.
- C. Display the printed model in the hallway.
- D. Students memorize the labeled parts.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

In a project funded by a local garden club, students test whether fertilizer affects bean growth: group A gets no fertilizer on a sunny windowsill, and group B gets fertilizer in a dim corner. What is the main design flaw?

- A. A third fertilizer level is needed before any conclusion.
- B. Group A should also get fertilizer so conditions match.
- C. Light also differs, so any growth difference cannot be linked to fertilizer.
- D. The garden club funding may bias the results.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

5. [TRANSFER — reversed stem / different variable]

Students test whether temperature affects catalase activity by timing bubbles from hydrogen peroxide. Which should be held constant?

- A. The rate of bubbling
- B. Amount of enzyme, peroxide concentration, and pH
- C. Temperature
- D. Only the brand of test tubes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

6.

A teacher at a school with a 90-minute block schedule wants students to study how allele frequencies change over 100 generations of natural selection. What resource fits best?

- A. A population-genetics simulation that runs many generations quickly
- B. A video lecture on natural selection
- C. Breed fruit flies for 100 generations during the course
- D. Nothing digital; simulations never count as real science.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

7. [TRANSFER — reversed stem / different variable]

When is a hands-on wet lab the better choice over a simulation?

- A. When the process takes thousands of years
- B. Always, whatever the cost or danger
- C. Never; simulations are always cleaner
- D. When it is safe and doable in class time, and the goal includes lab skills or working with real data

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

8.

On a field trip to a state forest, students learn that a large oak gained about 500 kg over several decades. Most say the mass came from the soil. Which framing and answer are correct?

- A. Matter and energy: the mass came from sunlight.
- B. Matter and energy: most of the mass came from CO₂ in the air, fixed by photosynthesis.
- C. Matter and energy: most of the mass came from water alone.
- D. Structure and function: most of the mass came from soil minerals.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A pre-assessment on the school's new learning platform shows many students believe giraffes stretched their necks and passed the longer necks to offspring. What is the best instructional move?

- A. Have students test the idea against evidence (for example, whether acquired traits appear in offspring), then build a model of variation and differential survival.
- B. State the correct definition once and move on.
- C. Grade the pre-assessment strictly to motivate study.
- D. Skip the topic to avoid confusion.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Students swab classroom surfaces onto agar plates in a lab that recently added a second sink. Which procedure is correct?

- A. Incubate at 37°C to speed growth.
- B. Tape plates shut, incubate near room temperature, observe without opening, and sterilize before disposal.
- C. Open the plates to count colonies more easily.
- D. Throw plates in the regular trash, since classroom bacteria are always harmless.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Why are school cultures from environmental swabs incubated near room temperature instead of 37°C?

- A.** Agar melts at 37°C.
- B.** Colonies would appear too fast to observe.
- C.** 37°C favors microbes adapted to the human body, including pathogens.
- D.** Bacteria cannot grow at 37°C.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Before a preserved-specimen dissection in a lab with new stainless-steel trays, a student raises an ethical objection. What is the appropriate response?

- A.** Excuse the student from the objectives with no replacement.
- B.** Require the dissection, since all students must do identical work.
- C.** Allow the objection but lower the student's grade.
- D.** Offer an equivalent alternative, such as a virtual dissection, that meets the same objectives.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

~~In a room with a new fume hood,~~ students claim that yeast makes more CO₂ at 35°C than at 20°C. Their balloons measured 18 cm around at 35°C and 9 cm at 20°C. Which statement is the reasoning?

Target word: identify reasoning **Decides:** Reasoning connects data to claim with a principle.

A	Confusion set	That is the evidence.
B	Absolute language	Too much heat denatures enzymes.
C	Correct	Applies a scientific principle to explain the data.
D	Dropped element	Restates the claim; the principle is missing.

2. Answer D Concept 1 • transfer

A student writes: Claim, the plant grew toward the light. Evidence, the stems bent 40° toward the window. Reasoning, “because the stems bent toward the window.” What feedback is most useful?

Target word: critique the reasoning **Decides:** Reasoning repeats evidence.

A	Adjacent category	More data does not fix missing reasoning.
B	Wrong stem number	Precision is not the problem.
C	Inverted operation	Those parts are already correct.
D	Correct	Points to the missing scientific link.

3. Answer A Concept 2

A teacher prints a DNA model on a 3-D printer ~~bought with a district grant~~. Which use best reflects the practice of developing and using models?

Target word: model as a tool **Decides:** Models explain, predict, and have limits.

A	Correct	Uses the model to reason and names its limits.
B	Absolute language	Every model simplifies.
C	Camouflage	Reacts to the printer detail.
D	Adjacent category	Recall, not modeling.

4. Answer C Concept 3

~~In a project funded by a local garden club,~~ students test whether fertilizer affects bean growth: group A gets no fertilizer on a sunny windowsill, and group B gets fertilizer in a dim corner. What is the main design flaw?

Target word: design flaw **Decides:** Light differs as well as fertilizer.

A	Adjacent category	More levels help, but the confound is the main flaw.
B	Inverted operation	That removes the independent variable.
C	Correct	A second variable confounds the result.
D	Camouflage	Funding is not the design flaw.

5. Answer B Concept 3 • transfer

Students test whether temperature affects catalase activity by timing bubbles from hydrogen peroxide. Which should be held constant?

Target word: controlled variables **Decides:** Everything except temperature.

A	Confusion set	That is the dependent variable.
B	Correct	These could also change the rate.
C	Inverted operation	That is the variable being tested.
D	Dropped element	Drops the variables that matter.

6. Answer A Concept 4

A teacher ~~at a school with a 90-minute block schedule~~ wants students to study how allele frequencies change over 100 generations of natural selection. What resource fits best?

Target word: choose the resource **Decides:** 100 generations cannot be observed in class.

A	Correct	Makes an impossible timescale observable.
B	Adjacent category	Students do not generate or analyze data.
C	Dropped element	Drops the timescale; that would take years.
D	Absolute language	Scientists use models and simulations routinely.

7. Answer D Concept 4 • transfer

When is a hands-on wet lab the better choice over a simulation?

Target word: when real labs win **Decides:** Feasible, safe, and lab skill or real data is the goal.

A	Confusion set	That is when a simulation is needed.
B	Absolute language	Some processes are unsafe or impossible in class.
C	Inverted operation	Clean data hides real-world variation students need to handle.
D	Correct	Real labs build skills simulations cannot.

8. Answer B Concept 5

~~On a field trip to a state forest,~~ students learn that a large oak gained about 500 kg over several decades. Most say the mass came from the soil. Which framing and answer are correct?

Target word: CCC and source of mass **Decides:** Tracking matter; carbon comes from CO₂.

A	Inverted operation	Light supplies energy, not atoms.
B	Correct	Carbon from air makes up most of the dry mass.
C	Dropped element	Water adds hydrogen and oxygen but drops the carbon from CO ₂ .
D	Confusion set	Soil supplies only a small share of the mass.

9. Answer A Concept 6

A pre-assessment ~~on the school's new learning platform~~ shows many students believe giraffes stretched their necks and passed the longer necks to offspring. What is the best instructional move?

Target word: addressing a misconception **Decides:** Acquired traits are not inherited; students need conflicting evidence.

A	Correct	Conflict with evidence, then a better model.
B	Dropped element	Drops the evidence students need to change their thinking.
C	Adjacent category	Pre-assessments diagnose; they are not for grading.
D	Inverted operation	Leaves the misconception in place.

10. Answer B Concept 7

Students swab classroom surfaces onto agar plates ~~in a lab that recently added a second sink~~. Which procedure is correct?

Target word: safe culture handling **Decides:** Unknown microbes may be pathogens.

A	Inverted operation	Body temperature favors organisms that infect people.
B	Correct	Standard school culture safety.
C	Dropped element	Drops the rule to keep grown plates sealed.
D	Absolute language	Unknown cultures must be treated as potentially harmful.

11. Answer C Concept 7 · transfer

Why are school cultures from environmental swabs incubated near room temperature instead of 37°C?

Target word: reason for the rule **Decides:** 37°C is human body temperature.

A	Confusion set	Solid agar stays firm well above 37°C.
B	Adjacent category	Speed is not the safety concern.
C	Correct	Lower temperatures select against them.
D	Inverted operation	Many grow best there.

12. Answer D Concept 8

Before a preserved-specimen dissection ~~in a lab with new stainless steel trays~~, a student raises an ethical objection. What is the appropriate response?

Target word: response to objection **Decides:** Learning objectives stay; the method can change.

A	Dropped element	Drops the learning goals.
B	Absolute language	Many policies call for alternatives.
C	Adjacent category	Penalizing a valid alternative is unfair.
D	Correct	Respects the objection and keeps the learning goals.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Claim, evidence, reasoning	A claim answers the question, evidence is the data, and reasoning links the evidence to the claim with a scientific principle.	Reasoning that just restates the evidence or the claim.	A student argument with three parts to label or critique.
2	Developing and using models	Models are tools to explain and predict; students use them, test them against evidence, revise them, and name their limits.	Treating a model as an exact copy or a diagram to memorize.	A physical or digital model and a question about how to use it.
3	Controlled investigations	Change one independent variable, measure the dependent variable, and hold all other conditions constant.	Letting a second variable (light, temperature) differ between groups.	Two groups that differ in more than one way.
4	Simulation vs. hands-on lab	Use simulations for processes too slow, large, dangerous, or costly to run in class; use hands-on labs when feasible and when lab skill or real data is the goal.	Believing one is always better.	A process over many generations, or a lab-skills objective.
5	Crosscutting concepts in biology	Matter and energy tracks where atoms come from and go; plant biomass comes mostly from CO ₂ fixed by photosynthesis, while sunlight supplies energy, not mass.	Believing plant mass comes from soil or from sunlight.	“Where did the mass of the tree come from?”
6	Addressing misconceptions	Draw out students’ ideas, confront them with evidence that conflicts, and help students build a better model; telling alone rarely changes them.	Just stating the correct definition or avoiding the topic.	A pre-assessment reveals a common wrong idea.
7	Microbiology culture safety	Treat unknown cultures as potential pathogens: seal plates, incubate near room temperature (not 37°C), never open grown plates, and sterilize before disposal.	Incubating at body temperature or opening plates to inspect them.	Swabbing surfaces onto agar plates.
8	Dissection ethics and alternatives	Offer an equivalent alternative (virtual or model dissection) that meets the same objectives to students with ethical objections.	Forcing participation, penalizing the student, or dropping the objectives.	A student objects before a dissection.

Course scope (public description): Science Methods—Secondary Biology focuses on teaching methods specific to science for graduate students seeking an endorsement in secondary biology. This course focuses on the design and teaching of standards-based lessons using the three dimensions of science (science and engineering practices, crosscutting concepts, and disciplinary core ideas) and the appropriate integration of technology into those lessons. Students in this course work within their content areas to evaluate, enhance, and plan appropriate science instruction. This course includes laboratory safety training and certification, which includes safe laboratory

practices and procedures for science classrooms and the proper use of personal protective equipment. Previous coursework in curriculum, instruction, and assessment is a prerequisite for this course.